1. Perform a simple calculation (Relevance: Extremely High)

Analysis: Solving this problem ultimately requires calculation. After recognizing that this is a $45^\circ - 45^\circ - 90^\circ$ isosceles right-angled triangle, the student needs to calculate $150 \times \sqrt{2}$ or use $\sin(45^\circ) = \frac{150}{\text{Rope Length}}$ to arrive at the result of approximately $212\ \text{m}$

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7. Manipulate numbers, graphical and statistical data and information, algebraic expressions and equations, and geometric representations (Relevance: Extremely High)

Analysis: The problem provides a geometric representation (a right-angled triangle with height and angle). Students must process these **geometric representations**, extracting the abstract values ($150\ \text{m}, 45^\circ$) from the diagram and applying them to a mathematical model.

5. Understand and utilise constructs based on definitions, rules and formal systems including employing familiar algorithms (Relevance: High)

Analysis: Students must apply rules from a **formal system** (trigonometry or Pythagorean theorem). This includes understanding the definitions of side-angle relationships in a right-angled triangle and employing familiar algorithms (like $a^2 + b^2 = c^2$ or the ratio rules for special right triangles).

10. Make sense of, relate and use a variety of representations when interacting with a problem (Relevance: High)

Analysis: This problem requires students to relate the **text description** ("vertical height", "rope length") to the **graphical representation** (the diagram with angles and dashed lines). Understanding the relationship between these different representations is key to solving the problem.

13. Use an understanding of the context to guide or expedite the mathematical solving process, e.g. working to a context-appropriate level of accuracy (Relevance: Medium)

Analysis: The problem is set in a practical sailing context. Students need to understand the physical meaning of "vertical height" and "rope length". Furthermore, the question asks for the "approximate" length, guiding students to select the most appropriate approximate option based on context.

11. Switch between different representations in the process of finding solutions (Relevance: Medium)

Analysis: During the solution process, students need to convert visual information from the diagram (the triangle) into an algebraic expression or a trigonometric equation. This **switch** from graphical to symbolic representation is a necessary step in finding the solution.

1. Perform a simple calculation (相关性：极高)

分析： 解决此题最终需要进行计算。在识别出这是一个 $45^\circ - 45^\circ - 90^\circ$ 的等腰直角三角形后，学生需要计算 $150 \times \sqrt{2}$ 或使用 $\sin(45^\circ) = \frac{150}{\text{绳长}}$ 进行计算，得出约 212 m 的结果 (p. 1)。

7. Manipulate numbers, graphical and statistical data... geometric representations (相关性：极高)

分析： 题目提供了一个几何图形表示（直角三角形及其高度、角度）。学生必须处理这些**几何表示**，将图形中的抽象数值（150 m，45°）提取出来并应用到数学模型中 (p. 1)。

5. Understand and utilise constructs based on definitions, rules and formal systems... (相关性：高)

分析： 学生必须运用**形式系统**（三角学或勾股定理）中的规则。这包括理解直角三角形中边角关系的定义，并应用熟悉的算法（如勾股定理 $a^2 + b^2 = c^2$ 或特殊直角三角形的比例规则）(p. 1)。

10. Make sense of, relate and use a variety of representations when interacting with a problem (相关性：高)

分析： 该问题要求学生将**文字描述**（“垂直高度”、“绳长”）与**图形表示**（带有角度和虚线的示意图）联系起来。理解这些不同表示方式之间的关系是解决问题的关键 (p. 1)。

13. Use an understanding of the context to guide or expedite the mathematical solving process... (相关性：中)

分析： 题目设定在实际航海情境中。学生需要理解“垂直高度”和“绳长”在物理空间中的意义。此外，题目要求计算“大约”的长度，这引导学生根据上下文选择最合适的近似选项 (p. 1)。

11. Switch between different representations in the process of finding solutions (相关性：中)

分析： 在解题过程中，学生需要将图形中的视觉信息（三角形）转换为代数表达式或三角方程，这种从图形表示到符号表示的**切换**是寻找解决方案的必经步骤 (p. 1)。